

Lab Book Marking Scheme and Common Debugging First Steps

Version 2 – 17/09/19

Common Debugging First Steps

- Fuse Blown on Protoboard?
- Circuit constructed correctly?
 - Nodes connected correctly?
 - Right components?
 - Individual components working correctly?
 - Power on?
- Can you measure another quantity in your circuit to verify that the circuit is working correctly?
- Measurement equipment setup correctly
 - Voltage or timescales correct?
 - Reading right channel?
 - AC vs DC settings on oscilloscope or DMM?
- Do you understand what you are measuring?
- Do you have an alternative method/technique for comparison?
 - Simulations
 - How long will it take to create and run a simulation?
 - Theoretical calculations
 - Alternative instrument measurement

Lab Book Marking Scheme

- Each section marked out of 5. Prepare lab book ahead of time

General organization

- Referencing instruction on OnQ
- File names and locations for sims or important documents, describe what is in the file
- Dates on all pages
- Titles to sections, even for roughwork and calculations
- Table of contents up to date
- Pages numbered
- Use pen, even when graphing

Graphs/Tables

- Titles
- Axis labels
- Drawn neatly
- Ruler used
- Short description of graph/table contents

Circuit Diagrams

- Neatly drawn
- Proper symbols used
- Titled/labeled
- Values of components in circuit
- Sources used for circuit?
 - Function gen?
 - Protoboard?
 - Battery?
 - Other?
- Where did you get special/irregular components?

Equipment used during lab

- Note name, make, model, instrument type used in each lab
- User manual file names and locations
- Reference page where you introduced the instrument if the same every lab

Commentary/ Debugging comments/ Questions from lab instructions

- Neat writing – can someone else read it?
- Clearly answers questions
- Includes commentary on troubleshooting equipment
- Not verbose – comment on serious time-consuming debugging efforts
 - Say what you have tried and why it did not work
 - Log steps so you do not repeat yourself – process of elimination
- Reference figures and tables with labels and page numbers

At the end of each session:

If you did not finish the lab: Record -

- Why you did not finish
- What was completed
- What is left to do in the lab/ project
- How long you worked during the session

If you did finish the lab: Record –

- That you completed the lab
- How long it took you to complete the lab/ work